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Surface plasmon based plasmonic sensors: A review on their past, present and future

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ABSTRACT

Surface plasmon-based plasmonic sensors have witnessed special attention in the research field due to their small size, ease of implementation, and remote sensing application. In terms of sensitivity, tunability, ultra-compactness, and effective light confinement, the plasmonic sensor has outperformed traditional optical sensors. Noble metals like gold, silver, and copper are used to stimulate surface plasmons and provide high sensitivity. The wavelength and angular interrogations are two types of interrogations that are commonly used to measure sensitivity. By combining nanostructured materials such as graphene, black phosphorous (BP), and Molybdenum disulfide (MoS2) with noble metals, the sensor's sensitivity has been enhanced. These catalytic type sensors can be used to sense the physical, chemical, and biological parameters. This study reviews the progress and developments of plasmonic sensors. Furthermore, the performance and application of various plasmonic approaches are briefly summarized. This review analyzes in detail about different topologies of surface plasmon-based plasmonic sensors, metals and nanomaterials that can be utilized in plasmonic sensors to enhance their sensitivity and the applications, emerging trends, challenges, and future scope of plasmonic sensors.

1. Introduction

A photonic crystal (PhC) is a periodic structure that exhibits periodicity in a specific direction or has a different dielectric constant. For regulating light flow, the PhC structure has been classified into three types: one, two, and three-dimensional structures. Two-dimensional photonic crystals are remarkable for research and commercial applications (Takano et al., 2006; Christopher et al., 2015). One of the unique properties of these structures is a forbidden frequency spectrum called a photonic bandgap (PBG) (Joannopoulos et al., 2007). Parameters such as the refractive index of the dielectric component, lattice parameters, and the crystal structure depend on the bandgap properties. PhC sensors have been widely used in medical, chemical, environmental, and food detection in recent (Jiong-Ju et al., 2020).

Surface plasmons (SPs) refer to the coherent oscillations of conduction electrons on a material surface under the stimulation of quantized energies, i.e., photons, electrons, and phonons (Luo et al., 2013). Typically, there are two forms of SPs: the localized surface plasmons (LSPs) and the propagating surface plasmon polaritons (SPPs), which are the fundamental elements in plasmonic sensors (Choi and Choi, 2012). When a plasmonic nanomaterial is irradiated with an electromagnetic (EM) wave with a wavelength longer than the size of the nanomaterial, electron oscillations occur on the surface of the plasmonic nanostructure. Although localized surface plasmons resonance (LSPR) occurs when the electric field of incident light resonates with the collective oscillations of surface electrons. As a result, the nanostructure absorbs light energy. Resulting in a highly localized amplification of the EM field within a few nanometers of the nanostructure (Li et al., 2015). SPPs are propagating oscillations of surface electrons on thin metal sheets, unlike LSPs. When a p-polarized light enters the metal-dielectric surface at an angle greater than the critical angle, the total reflection phenomenon occurs, and the evanescent wave has generated on the metal surface. Free electrons on the metal surface are excited by evanescent waves to form surface plasmon waves (SPW). Surface plasmon resonance (SPR) will occur when the incident light's wave vector matches that of SPW, resulting in a large EM field on the surface of the metal. Thus, reflected light intensity is significantly reduced (Zeng et al., 2017). The theory of SPR and LSPR-based sensing is that the state of the resonance is highly sensitive to the surrounding medium's refractive index (RI).

Numerous SPR-based devices, such as plasmonic demultiplexers (Rakhshani Mansouri-Birjandi, 2016a), splitters (Hu et al., 2016), filters (Zhuang et al., 2015), and sensors (Rakhshani Mansouri-Birjandi,

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Fig. 1. (a) Square resonator (Ag nanorods in square array) (Rakhshani and Mansouri-Birjandi, 2017), (b) Hexagonal resonator (hexagonal array of nanoholes in the metal substrate) (Rakhshani and Mansouri-Birjandi, 2018).

2016b; Padhy et al., 2016; Bahramipanah et al., 2014), have been proposed and widely implemented. Compared to other types of traditional sensors, the SPR sensor has many advantages. The SPR sensor provides, above all, very high sensitivity, quick response, and smaller sensing region size.

A plasmonic sensor uses the change of the plasmon's spectral properties to serve as a transducer of the sensing signal. The effective refractive index (RI) of the structure can be changed by introducing different materials into the structure and leading to change in the plasmon properties, which in this way we can calculate the sensitivity of the structure. The effectiveness of these plasmonic sensors is evaluated based on certain characteristic variables, such as sensitivity, resolution, and signal-to-noise ratio (SNR). Sensitivity is the main parameter to consider in a sensor device's design. Using a slit waveguide coupled with a nanodisk resonator (Dolatabady et al., 2013), a ring resonator (Wu et al., 2014a), a hexagonal cavity (Rakhshani Mansouri-Birjandi, 2016c), a slit cavity (Zhu et al., 2012), and a single defect (Wu et al., 2014b), multiple plasmonic RI sensors have recently been proposed. In addition, various types of high dielectric constant materials, such as silver (Ag) and gold (Au), are an ideal choice because they help both localized and propagating surface plasmon modes (Moroz, 1999).

This paper discusses the basic concepts, recent breakthroughs, and applications of surface plasmon-based plasmonic sensors. We also summarized the various topologies and sensing techniques used in the plasmonic sensor. The importance of nanomaterials and the effects of plasmonic metals employed in the sensor are briefly reviewed. The structure of this article is as follows. Following the introduction chapter, Section 2 summarizes the architectures of commonly used plasmonic sensors. The various materials used in plasmonic sensors are discussed in Section 3. The current state of plasmonic sensor applications is discussed in Section 4. In Section 5, we end up our discussion of surface plasmon-based plasmonic sensors' present state, challenges, and future research directions.

2. Surface plasmon-based plasmonic sensor structures

Plasmonic sensors use different structures, including metal insulator metal (MIM), metal-dielectric-metal (MDM), silicon-on-insulator (SOI), and optical fiber. The sensor's structure plays a vital role in its efficiency. The efficiency of the sensor is influenced by structural parameters such as the radius of the rod/air hole, lattice value, and metal thickness. The different structures of the plasmonic sensor have discussed in the following sections.

2.1. Nano array resonator based plasmonic sensors

Most of the plasmonic sensors with a high sensitivity to the refractive index are composed of optical waveguides and a resonator. *Rakhshani* et al. proposed a plasmonic refractive index sensor. And it consists of two metal insulator metal (MIM) waveguides coupled with a silver nanorod array embedded into a square resonator. He used a silver Nanorod for unique optical properties of good confinement and low propagation losses. Long waveguides along the entire sidewalls of the resonator are the main advantage compared to the other conventional sensors because it gives easy light confinement between the resonator and waveguides. Detecting material is placed into the square resonator, resonance wavelength depends upon the refractive index of the detecting material. Finite-difference time-domain (FDTD) method had used to analyze the structure performance. The sensitivity of the structure was 2320 nm/RIU (refractive index unit). The sensor was used to identify the human blood group. Fig. 1(a), shows a square resonator, Silver (Ag) nanorods in a square array, and two waveguides. The gaseous sample diffused into the resonator based on the gas diffusion force in the vacuum circumstance. The liquid sample filled up the resonator using the

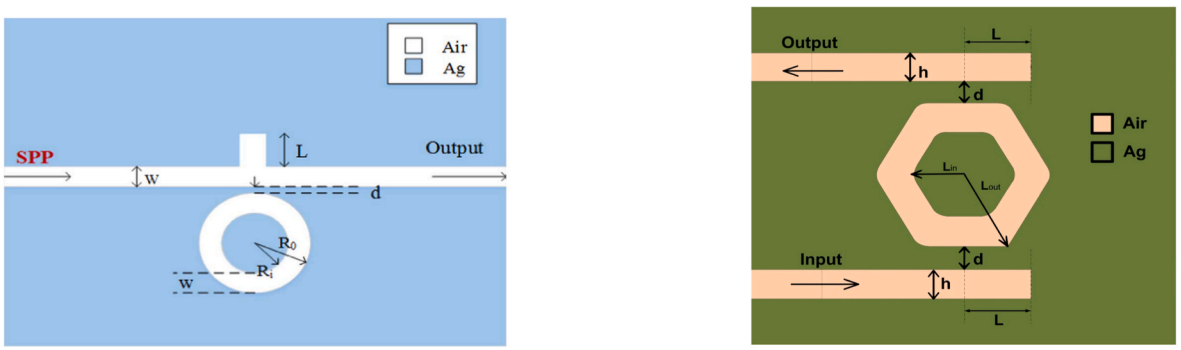


Fig. 2. (a) Ring resonator (Ring cavity resonator in Ag substrate) (He Yang, 2016), (b) Hexagonal cavity resonator (hexagonal resonator in Ag substrate) (Rakhshani and Mansouri-Birjandi, 2016).

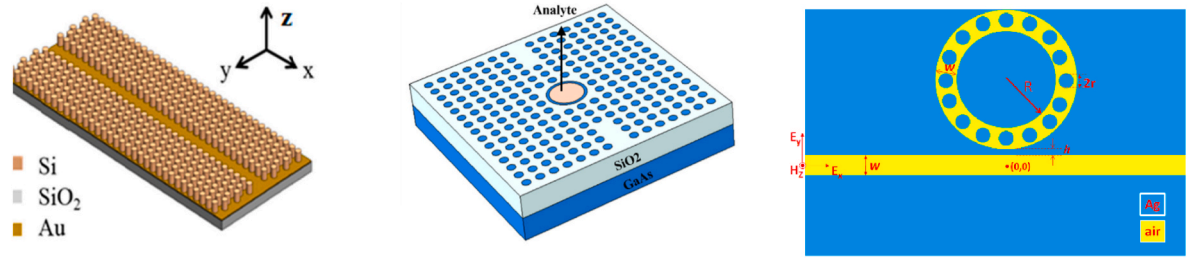


Fig. 3. (a) Hybrid waveguide composed of Si pillars on SiO2 substrate and embedded in a thin layer of gold (Jannesari et al., 2018), (b) Ring shaped defect coupled with waveguides (Rahman-Zadeh et al., 2019), (c) Nanoring containing Ag nanorods coupled with MIM waveguide (Chou Chao et al., 2020). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

nano-filling technique based on capillarity attraction (Rakhshani and Mansouri-Birjandi, 2017).

Rakhshani et al. proposed a plasmonic sensor. This sensor consists of two MIM waveguides coupled with an array of hexagonal nanoholes in a metal substrate (Ag). Nanoholes are very useful for designing compact plasmonic sensors with high sensitivity. This structure is used as a sensor to detect different human blood groups. An analyte was placed inside the nanoholes, RI of the nanoholes changed. And which leads to a change in the resonance wavelength of the structure. FDTD method was used to analyze the structure performance. And the sensitivity was 3172 nm/RIU. Fig. 1(b), a hexagonal array of nanoholes in the metal substrate and two waveguides coupled with this resonator. Compared with traditional square resonators, the hexagonal resonator can achieve a high value of sensitivity by decreasing radiation loss of SPP propagation (Rakhshani and Mansouri-Birjandi, 2018).

2.2. Ring resonator based plasmonic sensors

Plasmonic sensors have an oscillating nature and, under specific resonance conditions, tend to propagate on metal surfaces. This will allow for the detection of small changes in the refractive index.

Jialing He et al. presented a compact plasmonic coupled resonator structure. This sensor was composed of a ring resonator and a tooth-like resonator (stub) with a MIM waveguide. The finite element method (FEM) had been used to analyze the structure performance. The sensitivity of the sensor is 1100 nm/RIU. This sensor was used as an on-chip nanosensor. Fig. 2(a) consists of a ring plasmonic resonator and a stub located on both sides of the bus waveguide in Ag substrate (He Yang, 2016). *Rakhshani* et al. reported a 2D plasmonic waveguide and hexagonal ring cavity sensor. The proposed structure consists of a hexagonal cavity and two waveguides in the silver substrate. The sensing material had filled in the cavity, and the refractive index of the cavity will change. And this leads to the wavelength shift at the output channel. Sensing performance is numerically investigated and analyzed by the FDTD simulation method. The sensitivity of the structure is 4270nm/RIU. This structure had been used as a refractive index and temperature sensor. Fig. 2(b) consists of a hexagonal cavity resonator coupled with two air waveguides in the silver substrate. Unlike conventional ring resonators, hexagonal resonators have the advantage of a long adjacent interaction length along the entire sidewalls of the resonator. This powerful restriction on the separation of the distance between the resonator and the side-coupled waveguide can be eased (Rakhshani and Mansouri-Birjandi, 2016). *Abbas Alipour* et al. establish a dual-band tunable metal-dielectric-metal (MDM) structure for sensing applications. The structure was composed of three layers such as silver, a dielectric (SiO2), and silver respectively. The sensitivity of this structure was 1240.8 nm/RIU (Alipouret al, 2020).

In 2018, *M.A. Butt* et al. Presented a RI sensor based on a hybrid plasmonic waveguide and MIM microring resonator. In one single ring resonator, there are two slot hybrid plasmonic rings. The outer slot was formed by the Si-air-gold (Au) ring combination, and the inner slot was formed by the Au ring-air-Au disc. Gold had been preferred because of its biocompatibility and resistance to oxidation compared to Au. The numerical investigation had carried out by the FEM method. The sensitivity of these structures was 800nm/RIU. This sensor had been used for biomedical sensing applications (Butt et al., 1080). In 2019 he proposed another plasmonic sensor consisting of a square ring resonator coupled with two waveguides in a layer of silver deposited on a SiO2 substrate. Silver had been used in this sensor because of its low absorption and low power consumption. The sensitivity of these structures was 1367 nm/RIU. This sensor used on-chip nanosensors (Butt et al., 2019a).

2.3. Defect based plasmonic sensors

By introducing the defects inside the PhC structure, it is possible to localize the light in the PBG region. Defects are classified into two types: (1) A point defect or micro-cavity, and (2) a line defect or waveguide. Using the photonic bandgap effect, a lot of active and passive optical devices based on photonic crystals were designed, which may ultimately pave the way for photonic integrated circuits (PICs).

Reyhaneh jannesari et al. proposed a sensor based on 2D photonic crystal, silicon pillars embedded in a SiO2 substrate coated with a thin layer of gold in a hexagonal lattice. Creating a waveguide by removing one line of pillars. This structure is used for on-chip sensing applications. Fig. 3(a) shows the line defect, a hybrid waveguide composed of silicon (Si) pillars on silicon dioxide (SiO2) substrate and embedded in a thin layer of gold (Jannesari et al., 2018). Rahman-Zadeh et al. proposed a silicon-on-insulator (SOI) structure. In this structure, a 2D photonic crystal consists of a rectangular array of gallium arsenide (GaAs) rods in a SiO2 substrate. Cavities are ring-shaped defects accompanied by three rods on each side of the Ring, and it is coupled to the waveguide. The structure is suitable for lab on chip application. Inject a sample into the cavity, and the refractive index of the sensor was changed. This had a linear relationship between resonance wavelengths. The sensitivity of the structure was 739 nm/RIU.Fig. 3(b) shows the point defect Ring-shaped defect coupled with side waveguides (Rahman-Zadeh et al., 2019).

Chung-Ting chou Chao et al. designed a tunable refractive index-based plasmonic sensor. This sensor consists of a MIM waveguide connected with a nano ring containing Ag nanorods, which have a significant impact on the structure's sensitivity. The sensitivity of this design was 2080 nm/RIU. Silver nanorods were added to the nanoring to enhance sensitivity. Fig. 3(c) shows the Ag nanorods resonator (Chou Chao et al., 2020).

2.4. Fiber based plasmonic sensors

Photonic crystal fibers (PCF) are a new type of fiber that has a flexible structure and various unique features like high birefringence, endless single-mode, low transmission loss, and simple filling of material in fiber (Yanget, 2016; Chenget, 2013; Wong et al., 2012). PCF has been

Table 1
Comparison of the characteristics of the proposed plasmonic sensor structures reported in the literature.

Topology	Features	Setup	Stimulation	Wavelength Sensitivity (nm/RIU)	Detection range (Refractive index)	Year	Reference
Hexagonal ring cavity	Hexagonal cavity and two waveguides in Ag substrate replaced by air.	MIM	FDTD	4270	1–1.2	2016	Rakhshani and Mansouri-Birjandi (2016)
Square resonator	Silver nano-rods in a square Resonator.	MIM	FDTD	2320	1–1.1	2017	Rakhshani and Mansouri-Birjandi (2017)
Hexagonal resonator	Silver nanoholes in hexagonal substrate.	MIM	FDTD	3172	1–1.4	2018	Rakhshani and Mansouri-Birjandi (2018)
Concentric Ring Resonators	Waveguide coupled with a concentric double ring resonator in Ag substrate.	MIM	FDTD	1060	1.33–1.43	2018	Zhang et al. (2018)
Si ring resonator	Waveguide coupled with a ring resonator in Ag substrate and a silicon ring is placed inside the ring resonator.	MIM	FDTD	636	1.33–1.52	2019	Danaie Ali (2019)
Ring resonator	Ring resonator and stub coupled with waveguide surrounded by Ag substrate.	MIM	FEM	1100	1–1.02	2016	He Yang (2016)
Square type split-ring resonator	Waveguide with a baffle coupled with a circular split-ring resonance cavity	MIM	FEM	1700	1–1.1	2018	Yi et al. (2018)
Nano ring resonator	Waveguide-coupled nanoring containing silver nanorods	MIM	FEM	2080	1.14–1.61	2020	Chou Chao et al. (2020)
Square ring resonator	Square ring coupled with two waveguides in gold (Au) layer deposited in sio_2 substrate.	MIM	FEM	1367	1–1.2	2019	Butt et al. (2019a)
Square ring cavity	Multichannel MIM waveguide coupled with 3 square ring cavities in gold (Au) substrate.	MIM	FEM	1948.67	1–1.3351	2020	Butt et al. (2020)
Dual band Resonator	Silicon dioxide (SiO2) which sandwiched between the two silver thin films.	MDM	FDTD	1240.8	1.41–1.61	2020	Alipouret al (2020)
Grating coupled SPP sensor.	High indexed chalcogenide arsenic selenide grating embedded in a transparent substrate. A thin layer of gold placed next to the grating.	Metal-Analyte	FDTD	1133	1.33–1.345	2020	Joseph et al. (2020)
Side coupled ring resonator	Ring shaped defect coupled with waveguides	SOI	FDTD	739	1.33–1.5	2019	Rahman-Zadeh et al. (2019)
Nano cavity resonator	Thin gold metal layer is being embedded to the silver layer to attain higher.	SOI	FDTD	300	1.1–1.4	2019	(Chandra et al., 2019)
SPR based RI sensing probe	ITO layer and polymer matrix are used in the multimode silica glass fiber cladding to enhance sensitivity.	Optical fiber	FEM	10280	1.33–1.38	2021	Mishra et al. (2021)
SPR based fiber optic sensor.	ITO and silver (Ag) stripes are arranged alternatively to form a grating in the core. A thin layer of silicon is used to protect metals from oxidation.	Optical fiber	FEM	25000	1.33–1.38	2022	Mishra et al. (2022)
Dual core plasmonic sensor.	Dual core SPR based PCF sensor, gold is used as a plasmonic material to enhance sensitivity.	PCF	FEM	10700	1.39–1.40	2020	Shafkat et al. (2020)
Dual core D-shaped SPR based biosensor.	D-shaped six-fold PCF bio sensor, silver nanowire embedded to enhance the sensitivity.	PCF	FEM	4000	1.35–1.50	2021	Chu et al. (2021)

extended to the plasmonic sensor by filling/coating a noble metal such as gold (Shafkat et al., 2020), silver (Chu et al., 2021), copper (Azman et al., 2019), and titanium (Jabin et al., 2019) in the PCF structure. Surface plasmon (SP) develops at the metal/dielectric interface when plasmonic materials are exposed to electromagnetic waves due to the collective oscillations of conduction electrons (Heikal et al., 2013). At the phase matching condition, the core guided mode transfers maximum energy to the plasmon mode. As a result of coupling, core mode energy decreases, and confinement loss increase. This phenomenon is known as Surface Plasmon Resonance (SPR) (Burstein et al., 1974).

Satyendra Kumar Mishra et al. proposed an SPR-based refractive index sensing probe. This silica glass multimode fiber probe uses an

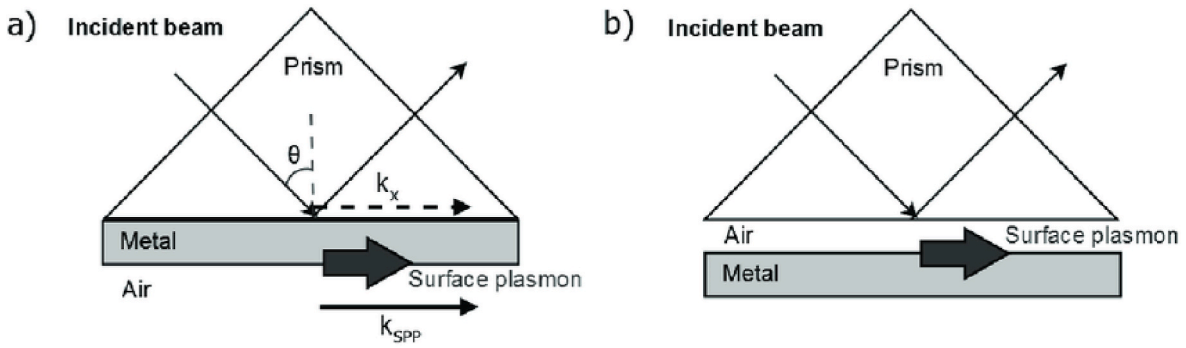


Fig. 4. (a) Kretschmann configuration of an SPR sensor (Kretschmann Reather, 1968), (b) Otto configuration (Otto, 1968).

indium tin oxide (ITO) layer and polymer matrix on the cladding. The ITO induces the plasmonic effect and the polymer layer enhances the sensitivity of 10280 nm/RIU in the RI range of 1.33–1.38 (Mishra et al., 2021). In 2022 *Satyendra Kumar Mishra* et al. designed an SPR-based fiber optic sensor. In this fiber, indium tin oxide (ITO) and silver (Ag) stripes are arranged alternatively to form a grating. A thin layer of silicon is used to protect metals from oxidation as well as improve sensitivity. The maximum sensitivity of 25000 nm/RIU was achieved in the RI range of 1.33–1.38 (Mishra et al., 2022). *Araf Shafkat* et al. reported an SPR-based PCF sensor with a sensitivity of 10700 nm/RIU for the detection range 1.39–1.40. This sensor gold is used as a plasmonic material to stimulate the plasmonic effect (Shafkat et al., 2020). Table 1 listed the recent research related to plasmonic sensor structures.

3. Materials used in plasmonic sensors

3.1. Metals

Surface plasmon waves (SPWs) are created on the metal-dielectric interface when a monochromatic light source is applied directly to the metal surface. However, the resonance condition is not satisfied due to a lack of light accumulation on the metal surface. The metal surface is coupled to a prism, allowing it to collect all of the light. *Otto* et al. (Otto, 1968) suggested coupling the metal with the prism. He presented a configuration shown in Fig. 4 b. The metal layer is located at the base of the prism. The air gap between the metal layer and the base prism is retained. On the air–metal interface, the SPW is created. The associated evanescent wave (EW) from the air to metal decreases exponentially.

It is hard to sustain an air gap Kretschmann et al. (Kretschmann Reather, 1968) improved the Otto configuration shown in Fig. 4 a. In this configuration, the metal layer is deposited on the prism base. Therefore, there is no gap between metal and prism. That is a beneficial configuration for enhancing the sensitivity of the SPR sensor. The related EW in the metal and dielectric media decreases exponentially. The dielectric layer is coupled with a sensing medium or analyte for detecting the biomolecules. Metals such as gold (Au), silver (Ag), copper (Cu), aluminum (Al), indium (In), and sodium (Na) are capable of producing SPs when there is light on the surface of the product. These metals have a high charge-carrier concentration and light absorption in the visible range, which can improve the plasmonic effect. The condition is referred to as the SPR condition when the wave vectors of incident light and SPs match. The change in the RI of the medium at the SPR sensing surface can be measured by the SPR sensor. The properties such as wavelength, phase, dielectric constant, and angle of the sensing medium will be adjusted accordingly when the RI changes.

Refractive index-based SPR sensors, i.e., gold, copper, or silver planar films, have been used to detect the binding of analytes on or near a metal surface. The two most widely used SPR sensor substrate materials are Au and Ag. Ag substrate-based SPR sensors display greater detection sensitivity compared with Au substrate. However, Ag's low chemical stability in SPR sensors hampers its application prospects. (Szunerits et al., 2013).

The metal is composed of a complex dielectric constant (n + ik). Reflection and absorption are described by the real part (n) and an imaginary part (k) of the dielectric constant. The SPR spectrum also depends on the dielectric constant's real and imaginary parts.

The Drude–Lorentz Model is helpful to determine the optical response and RI of the Ag or Au (Cao et al., 2012):

$$n^2 = (1 - \lambda^2 \lambda c / \lambda p^2 (\lambda c + i \lambda))$$

Where λ, λc, and λp denote the operating wavelength of the monochromatic source of light, collision wavelength, and plasma wavelength, respectively. The values of λc and λp for Ag are 176.14 nm and 145.14 nm, while for Au, they are 8934 and 168.26 nm, respectively. Because it is feasible to increase optical nonlinearity at a higher frequency, the wavelength 633 nm is chosen as the operational wavelength for each material. The corresponding refractive index values for the various metals are shown in Table 2.

Table 2
Refractive index for the different plasmonic metals and Nanomaterials.

Metal/ Nanomaterial	Wavelength (nm)	Refractive index (n + ik)	Reference
Silver (Ag)	633	0.056253 + 4.2760i	(Karki et al., 2022), (and McCarthy, 1975)
Gold (Au)	633	0.1838 + 3.4313i	(Maurya et al., 2015), (and McCarthy, 1975)
Copper (Cu)	633	0.27002 + 3.4081i	(Karki et al., 2022), (and McCarthy, 1975)
Platinum (Pt)	633	0.23 + 4.07i	Ordal et al. (1983)
Graphene	633	3.0 + 1.1487i	Maurya et al. (2015)
Black Phosphorus (BP)	633	3.5 + 0.01i	Wu et al. (2017)
Blue Phosphorus (BlueP)	633	2.1666 + 0.1005i	Li et al. (2020)
Molybdenum disulphide (MOS2)	633	5.9 + 0.8i	Maurya et al. (2015)

Rakhshani et al. reported an air-silver (Ag) based plasmonic sensor. Ag is employed because of its electromagnetic response. As a result, its power consumption is low compared to Au and Al. The sensitivity of the structure was 4270 nm/RIU (Rakhshani and Mansouri-Birjandi, 2016). *Chandra Shekar* et al. Presented a three-layer SPR sensor. SiO2 has been sandwiched between the Si substrate and the gold layer analyte has been placed upon the gold layer. This structure is used to detect the affected area with a particular set of tissue. The refractive index of normal cancer tissue and the refractive index of Melanoma cancer tissue were studied. The sensitivity of the structure is 300nm/RIU (Chandra et al., 2019). *M. A. Butta* et al. proposed a square ring cavity resonator. In this structure layer of silver is deposited in the SiO2 substrate air as an insulator. The structure gives a sensitivity of 1367 nm/RI U (Butt et al., 2019a).

3.2. Nanomaterials

Some other nanomaterials are used based on the Kretschmann configuration to increase the sensitivity of the SPR sensors. To increase the sensitivity of SPR biosensors, 2D nanomaterials such as graphene, black phosphorous (BP), and transition metal dichalcogenides (TMDCs), i.e., MoS2, Molybdenum diselenide (MoSe2), Tungsten disulfide (WS2), and Tungsten diselenide (WSe2), have attracted great attention (Kretschmann Reather, 1968). 2D nanomaterials have excellent thermal conductivity, optically transparent mechanical conductivity, high surface-to-volume ratio, and high elasticity. The high surface area to volume ratio helps to improve the sensitivity, quality factor, and detection accuracy of SPR sensors. Among these, graphene is the nanomaterial most widely used because it has a high surface-to-volume ratio, fast electron mobility, graphene layer balanced structure, indirect bandgap, highly optically transparent (97.7%), etc. (Zhou et al., 2020). In most cases, the efficiency of plasmonic sensors is enhanced by using 2D nanomaterials. The enhancement mainly depends on the following features, 2D nanomaterials bind to proteins and nucleic acids.2D nanomaterials, such as graphene, Transition metal dichalcogenides (TMDs), and hexagonal boron nitride (h-BN), can resist the oxidation of the active plasmonic metal. Next, the addition of a 2D nanomaterial layer could increase the electrical field at the interface and accelerate the transfer of charges between plasmonic metal and probe molecules due to the high mobility of the carrier, and thereby improve the sensitivity of the plasmonic sensors (Zhou et al., 2020). The refractive index and dielectric constant values for the various 2D nanometals are listed in Table 2.

Zhitao lin et al. proposed a modified structure of a four-layer SPR sensor by using graphene and MoS2.In this structure, MoS2 is

Table 3
Summary of the 2D nanomaterials used in plasmonic sensor reported in the literature.

Configuration	Sensitivity		Detection Range (Refractive index)	Year	Reference
	Wavelength Sensitivity (nm/RIU)	Angular sensitivity (deg/RIU)			
Ag/air/Ag	4270	–	1–1.2	2016	Rakhshani and Mansouri-Birjandi (2016)
Ag/Si/air	636	–	1.33–1.52	2019	Danaie Ali (2019)
Ag/air/Au	1367	–	1–1.2	2019	Butt et al. (2019a)
Ag/SiO2/air	2080	–	1.14–1.61	2020	Chou Chao et al. (2020)
Au/Si/SiO2	300	–	1.1–1.4	2019	Chandra et al. (2019)
Au/Cr/MoS2	2793.5	–	1.333–1.360	2019	Chen et al. (2019)
Au/air/Au	1948.67	–	1–1.3351	2020	Butt et al. (2020)
ITO/Si/SWCNT	9780	–	1.33–1.335	2017	Mishra et al. (2017a)
Cu/SiO2/Polymer	250000	–	1.5375–1.5515	2016	Mishra et al. (2016)
Au/MoS2/Au	–	182	1.33–1.335	2016	Lin et al. (2016)
Au/MoS2/graphene/Ni	–	229	1.33–1.335	2019	Nisha et al. (2019)
Ag/Graphene/MoS2	–	73.5	–	2018	Aksimsek et al. (2018)
Ag/BP/MoS2	–	432.15	1.32–1.36	2018	Anuj et al. (2018)
MgF2/Rhodium/graphene	–	240	1.33–1.36	2017	Mishra et al. (2017b)
Air/MoS2/Al/MoS2/graphene	–	190.83	1.33–1.42	2016	Wu et al. (2016)

Sandwiched between two gold films. The angular sensitivity of the sensor is 182 deg/RIU (Lin et al., 2016). *Leiming Wu* et al. reported a heterostructure plasmonic sensor. MoS2 -Graphene heterostructure enhance the sensitivity of the SPR sensor. Compared to graphene, MoS2 has a high imaginary part of the dielectric constant and, this means that MoS2 can absorb moonlight energy. The angular sensitivity of the structure is 190.83 deg/RIU (Wu et al., 2016). *Shushama* et al. proposed a seven-layer modal SPR biosensor with a sensitivity of 210 deg/RIU (Nahar Shushama et al., 2017). *Sinan Aksimsek* et al. presented graphene- MoS2-Silver (Ag) hybrid structure under minimum light intensity. Graphene is used as bio recognizing element. MoS2 is deposited between graphene and metal. The sensitivity was 2.3 times larger than the conventional SPR sensor (Aksimsek et al., 2018). *Anuj. k. Sharma* et al. proposed a four-layer SPR sensor. The use of a Blue Phosphorene (BP) –MoS2 heterostructure layer on top of an Ag substrate to prevents the metal from oxidizing. This arrangement achieves a sensitivity of 432.15 deg/RIU (Anuj et al., 2018).

The SPR sensor based on Chromium (Cr)– Au– MoS2 is reported by *Yaofei Chen* et al. MoS2 nanosheets are deposited on the SPR Sensor's surface. The number of MoS2 deposited cycles determines the sensitivity. In this structure, he used two cycles of MoS2 deposition. The wavelength sensitivity of the sensor is 2793.5 nm/RIU. Sensitivity was 30.67 percent enhanced compared without MoS2 (Chen et al., 2019). *Nisha* et al. proposed a MoS2-Graphene- Au- Nickel (Ni)-based five-layer SPR sensor. MoS2 layer sandwich between gold (Ag) and Ni film, adding graphene over Ni film. This structure enhanced the sensitivity of 229 deg/RI U (Nisha et al., 2019). *Qiu* et al. reported a photoelectrochemical aptasensor. The plasmonic effect between two-dimensional graphite-phase carbon nitride (g- C3N4)/Bismuth molybdenum oxide (Bi2MoO6) and Au was used to find the different concentrations of carcinoembryonic antigen (CEA) (Qiu et al., 2018). *Akhilesh. K. Mishra* et al. proposed an SPR-based fiber sensor. In this multimode fiber, a portion of cladding was removed and the layers of ITO, silicon, and highly doped SWCNT (Single-wall carbon nanotube) are deposited to enhance the sensitivity of 9780 nm/RIU (Mishra et al., 2017a). The materials used in the SPR sensor are summarized in Table 3.

4. Plasmonic sensor applications

4.1. Biosensor

Biosensing describes the detection of biological substances such as viruses, bacteria, DNA, and proteins, from micrometers (μm) to nanometers (nm) scale. Biosensing is also essential for identifying biomolecules, microorganisms, biological structures, etc. The plasmonic biosensor works based on the refractive index variation due to the binding of biomolecules on the sensor layer. *J. B. Maurya.* et al. designed a sensor for bacteria (Maurya et al., 2018). *Rakhshani* et al. explained the plasmonic sensor for human blood group identification (Rakhshani and Mansouri-Birjandi, 2017, 2018). *Sinan Aksimsek* et al. proposed an SPR-based sensor for single-molecule detection. Its sensitivity was 2.3 times larger than the conventional SPR sensors (Aksimsek et al., 2018). *Chandra Shekar* et al. explained the application of the SPR sensor for Melanoma cancer tissues (Chandra et al., 2019).

4.2. Chemical sensor

In several fields such as drug and food monitoring, environmental monitoring, medicine, and biotechnology, the investigation and identification of chemical and biochemical substances are important. Changes in the analyte's complexity are achieved directly by measuring the RI using an SPR sensor. The main focus of the applications was on changes in the RI of the sensing layer caused by the adsorption of the analyte molecules. *Rakhshani* et al. explained the Ag nanorods in square resonator SPR sensor to investigate the Glucose concentration in water (Rakhshani et al., 2018). Md. *Moznuzzaman* et al. have also developed a model of highly improved 2D material (graphene and MoS2)-based SPR sensor for formalin detection (Biplob Hossain et al., 2019). *Anuj K. Sharma* et al. and *M.A.Butt* et al. proposed a gas sensor (Anuj et al., 2018; Butt et al., 2019b). *M.A. Butt* et al. discussed the fundamentals of SPR chemical sensors and their current developments for future applications (Chandra et al., 2019). *Satyendra K. Mishra* et al. reported an SPR-based fiber optic sensor for chlorine gas sensing. 1 cm of the cladding is removed from the silica fiber in the middle. The unclad section was covered with 40 nm silver film first, then a thin film of indium oxide doped with tin oxide. The sensitivity of the sensor was influenced by the thickness and concentration of the metals (Mishra and Gupta, 2015).

4.3. Sensor for the study of the physical quantities

Plasmonic sensors are used to measure physical quantities such as temperature, pressure, humidity, etc., measurement of displacement, and angular position that are defined based on SPR sensitivity to the momentum of the light wave incident. For the creation of SPR sensing devices, physical process events in different optical sensing materials were also observed (Singh et al., 2020). *Rakhshani* et al. explained the SPR sensor for the application of temperature measurement with high sensitivity of 4270 nm/RIU (Rakhshani and Mansouri-Birjandi, 2016).

Table 4
List of Application of the 2D Plasmonic sensor.

Application	Technique	Sensitivity		Detection range (Refractive index)	Year	Reference
		Wavelength Sensitivity (nm/RIU)	Angular sensitivity (deg/RIU)			
Temperature	SPR	4270	–	1–1.2	2016	Rakhshani and Mansouri-Birjandi (2016)
Biomedical	SPR	250000	–	1.5335–1.5515	2016	Mishra et al. (2016)
Blood group	SPR	2320	–	1–1.1	2017	Rakhshani and Mansouri-Birjandi (2017)
Blood group	SPR	3172	–	1–1.4	2018	Rakhshani and Mansouri-Birjandi (2018)
Biochemical	SPR	800	–	1–1.03	2018	Butt et al. (1080)
Single molecule	SPR	73.5	–	–	2018	Aksimsek et al. (2018)
Glucose concentration	SPR	892	–	1.32–1.355	2018	Rakhshani et al. (2018)
Biomedical	SPR	739	–	1.33–1.5	2019	Rahman-Zadeh et al. (2019)
Melanoma cancer tissue	SPR	300	–	1.1–1.4	2019	Chandra et al. (2019)
Gas	SPR	1320	–	1–1.15	2019	Butt et al. (2019b)
Chemical	SPR	1948.1	–	1–1.3351	2020	Butt et al. (2020)
DNA	SPR	–	194	1.33–1.39	2020	Rahmana et al. (2020)
Bacteria	SPR	–	109.39	1.33–1.4	2018	Maurya et al. (2018)
Gas	SPR	–	432.15	1.32–1.36	2018	Anuj et al. (2018)
Biosensing	SPR	–	229	1.33–1.335	2019	Nisha et al. (2019)
Formalin	SPR	633	79	1.34–1.41	2019	Biplob Hossain et al. (2019)

Gokhan Topcu et al. proposed a plasmonic sensor for pressure measurement working range up to 160 MPa (Topcu et al., 2019). Table 4. Summarize the applications of 2D plasmonic sensor.

5. Future perspective of plasmonic sensors

Surface plasmon based plasmonic nanosensors have prompted researchers to investigate the field of plasmonic sensors due to their flexibility, small size, and high sensitivity. Researchers are fascinated in structurally modified plasmonic sensors because they provide enhanced efficiency than standard plasmonic sensors. However, the demand for high quality, on the other hand, introduces new challenges.

(i) Improvement in the analytic figures of merit such as sensitivity, selectivity, and dynamic range. Using a thin layer of 2D nanomaterials such as graphene, black phosphorus (BP), and Mos2 is coated with noble metals (e.g., gold, silver, copper) to improve the performance of plasmonic sensors since these metals are free from corrosion and oxidation. This catalytic sensor system is sensitive because it can resist the oxidation of the active plasmonic metal. After all, 2D nanomaterials can bind to proteins and nucleic acids. Next, the addition of a 2D nanomaterial layer could increase the electrical field at the interface and accelerate the transfer of charges between plasmonic metal and probe molecules due to the high mobility of the carrier, thereby improving the sensitivity of the plasmonic sensor.

(ii) High detection range; Plasmonic sensors were initially used to detect changes in the ambient atmosphere in the bulk refractive index. Plasmonic sensors are currently used to detect changes in the local refractive index caused by biomolecular binding. Plasmonic sensors have a sensitive and moderate detection range, which is insufficient for distinguishing small compounds. However, they are more realistic and easier to use. Implementing the phase-detection approach can increase the detection range.

(iii) Plasmonic sensor miniaturization is commonly used in biosensing, pharmaceutical screening, and chemical sensing. The miniaturized sensor extends the range of potential applications. Recent developments in nanotechnology have made it possible to fabricate large sensors using micro and nano-fabrication technologies to minimize sensor size and thereby increase throughput and decrease costs.

6. Conclusion

In this review, surface plasmon-based plasmonic sensors are summarized and compared. Most of the recent research work on plasmonic sensors are carefully reported. The basic concepts, recent developments, and applications of plasmonic sensors are presented in this review briefly. Various plasmonic sensor structures and the noble materials are used in that structures have been thoroughly studied. The wavelength and angular sensitivity of various structures are studied scrupulously and noted that the noble metals are used in them to provide better sensitivity. Nanomaterials have been widely used in the plasmonic sensors due to their inherent advantages over conventional materials. So, nanomaterials are now being utilized for biosensing and industrial applications. Nanomaterials are also meticulously reviewed for their various applications. The nanomaterials not only enhance the sensitivity but also protect metal layers from oxidation. The plasmonic sensor's sensitivity has increased tremendously by using a metal/nanomaterial catalytic hybrid sensor. SPR considerably intensifies the signal near a metallic surface when compared to the SPP approach. As a result, many plasmonic sensors use the SPR approach only. The hybrid plasmonic sensors with SPR technique provide 2.5 times higher sensitivity than the traditional sensors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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